26900147 - Scheme of Evaluation

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (Honours) /MGU-BBA (Honours) MGU-BCA (Honours)
REGULAR EXAMINATION October 2026 Fifth Semester
Discipline Specific Elective (DSE)-MG5DSEBCA300 –DIGITAL IMAGE PROCESSING
(2024 ADMISSION ONWARDS)

Duration: 2 hours Max Marks: 70

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create(C), Skill(S), Interest (I) and Appreciation (Ap)

Part A, Very Short Answer Questions, Answer all questions
Each question carries 2 marks

1. Explain image sampling and quantization. [U]/[CO1]

Image sampling (1 mark): Sampling converts the continuous spatial coordinates of an image into a finite set of discrete pixel locations. Image quantization (1 mark): Quantization converts the continuous intensity values of the sampled image into a finite number of discrete intensity or gray levels.

2. Differentiate between an analog image and a digital image. [An] / [CO1]

Any two relevant points of difference between an analog image and a digital image. 1 mark each

3. Analyze the effect of dilation on an object. [An] / [CO3]

Dilation is a morphological operation that expands or thickens foreground objects' boundaries in an image using a structuring element. — 1 mark. Can connect nearby objects, fill small gaps or bridge narrow breaks, depending on the structuring element — 1 mark.

4. Identify one application of hit-or-miss transformation. [U] / [CO3]

The hit-or-miss transformation is a morphological operation used to - detect a particular pattern or shape in a binary image - Detecting specific shapes or patterns. Detecting corners or particular pixel configurations- Recognizing predefined binary patterns. Any two relevant applications — 2 marks.

5. State any two characteristics of Rayleigh noise. [K] / [CO4]

Any two characteristics 2 marks- It follows a Rayleigh probability distribution. Its probability distribution is non-negative asymmetric, right-skewed. used to model certain types of noise encountered in radar and ultrasound imaging. Its statistical characteristics depend on the scale parameter of the distribution.

[2x5 = 10]

Part B, Short Answer Questions, Answer any **5** questions, Each question carries **6** marks

6. Explain different color models used in image processing. [U] / [CO1]

   Explain any three color models (RGB, CMY, CMYK, HSI…) 2 marks each

7 Fundamental steps in Digital Image Processing : Any six steps with short description (or listing any six and detailed explanation of any [U] / [CO1]

three) 6 marks -Image acquisition, Image Enhancement, Image Restoration, Color image processing, Wavelets Transform, Compression, Morphological processing, Segmentation, Representation and description, Recognition, Knowledgebase

8. Write short note on a) Image processing hardware b)Image processing software c)Mass storage 2 marks each for explanation [U] / [CO1]

9. State the objectives of histogram equalization. Explain how it improves the contrast of a low-contrast image. [U] / [CO2]

   Objectives of histogram equalization 3 marks, Histogram equalization improves its contrast through the following steps: Calculate the histogram: Determine the frequency of each intensity level in the image. Compute the cumulative distribution function (CDF): Calculate the cumulative sum of the normalized histogram values.Transform pixel intensities: Use the CDF to map the original intensity values to new values distributed across a wider intensity range. 3 Marks.

10. Explain the 2-D DFT and its inverse transform. Describe image representation in the frequency domain. 2-DFT (2 Marks) Inverse transform (1 Mark) Describe image representation in the frequency domain. (3 Marks) [An] / [CO2]

11. Analyze how opening and closing operations can be used for noise removal. 3 Marks each [An] / [CO3]

12. Analyze how Otsu's method determines the optimal thresholding . Definition 2 marks. Explain between class variance,and optimal threshold selection 2 marks Explain how the selected threshold provide effective foreground background separation 2 marks [An] / [CO4]

[6x5 = 30]

Part C, Essay Questions, Answer any **2** questions, Each question carries **15** marks

13. For two pixels P(x,y)and Q(s,t) calculate their Euclidean, D4 and D8 distances for given coordinate values. Definition 2 marks each. Demonstration using suitable examples 3 marks each [A] / [CO2]

14. Analyze the fundamentals of morphological image processing. Discuss its basic concepts, terminology, and major applications with suitable examples. [An] / [CO3]

    Definition 3 marks, basic set theory for morphological operations structuring element concept of hit miss fit, neighboring elements 6 marks, applications and Explanation- how structure or shape of object change in morphological transformations using different structuring elements 6 marks

15. Compare region growing and region splitting and merging techniques. Definition 5 marks comparison 10 marks [An] / [CO4]

[15x2 = 30]

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